Oxalate Stoichiometry

Student Written Instructions

Safety Precautions

- Chemical splash goggles, gloves and apron must be worn at all times.
- Be careful not to touch hot objects. Immediately place any burns under cold water and inform the teaching assistant.
- Perform the steps involving heating and pyrolysis of the solid oxalate well inside the student hood. Ensure that student hood is firmly seated against the rubber gasket.
- Most metal salts are toxic if ingested. Some, like the $Ni(NO_3)_2 \bullet 6H_2O$, are carcinogens. Wear gloves!
- Thoroughly wash hands with soap and water before leaving the laboratory room.

MATERIALS LIST

- 0.888 M oxalic acid, $H_2C_2O_4$
- iron(II) chloride tetrahydrate, $FeCl_2 \bullet 4H_2O$
- nickel(II) nitrate hexahydrate, $Ni(NO_3)_2 \bullet 6H_2O$
- manganese(II) sulfate monohydrate, $MnSO_4 \bullet H_2O$
- acetone, C_3H_6O

OBJECTIVES

After completing this project, you will be able to:

- write and balance simple chemical equations
- convert between grams, moles, volume, and molar mass
- calculate amount of reactant or reactants and/or product or products given amount of other reactants or products using the balanced chemical equation
- identify the limiting reagent in a reaction
- calculate the theoretical yield of a reaction and amount of unused excess reagent when given the amounts of reactants
- convert between actual yield, percent yield and theoretical yield

INTRODUCTION

Kidney stones are a common medical condition for not just humans, but also dogs, cats, and even iguanas.[1] Chemically, most kidney stones are calcium oxalate formed by the precipitation of calcium ions with oxalate ions ($C_2O_4^{2-}$) in the urine.[2] Oxalates block the absorption of important nutrients, such as calcium and iron, resulting in increased metal ion concentrations, and subsequent oxalate precipitation and kidney stone formation.[3] Surprisingly, increased consumption of calcium in the diet is not correlated with increased occurrence of kidney stones.[2] However, eating high content oxalic acid foods, such as spinach, nuts, rhubarb, and sesame seeds, does lead to more kidney stones.[4] According to Guinness World records, the heaviest kidney stone reported weighed 620 g (21.87 oz) and had to be surgically removed from the kidney of a man in Pakistan in 2008.[5] Many kidney stones are passed without incident. However, once stones reach 2 -3 mm in diameter a person will experience significant pain.

Are there any other reasons oxalates are important? **Yes!** From kidney stones and metal oxide nanoparticle formation to oxalate's role in joint inflammation and autism and to their use as pigments and as evidence of life on Mars, the importance of oxalates is evident!

Oxalates in Context

Metal Oxalate	Effect or Use
calcium oxalate	kidney stones in humans and animals[1-4]
transition metal oxalates	as precursors to metal oxide nanoparticles[6,7,8]: applications in catalysis[9] and semi-conductors[10]
iron(II) oxalate	used as yellow pigment: paints, glass, and plastic
naturally occurring metal oxalates, e.g. FeC_2O_4 (humboldtine)	product of fungi/lichen growth: presence used as evidence for life on other planets[11]
metal oxalate crystals	formation in bones/joints: associated with inflammation,[12] anemia, immunosuppression
low oxalate diet	as treatment for autism: preliminary research indicates that autistic children have high amounts of oxalates in urine[13,14]

Many metal ions, except for Group 1A, the alkali metals, form an oxalate precipitate at low oxalic acid concentrations. In human urine, the most abundant precipitating metal ion is calcium, which leads to the common form of kidney stones. This experiment will examine the precipitate formed by oxalic acid with other metals, iron, nickel and manganese.

SCIENTIFIC BACKGROUND

Vacuum filtration

Vacuum filtration is much faster and more efficient than gravity filtration. To perform vacuum filtration, first attach the vacuum flask to a ring stand using a clamp. Once the entire apparatus is set up, it will be top-heavy and prone to falling over. Place a Büchner funnel in the mouth of the clamped vacuum flask and attach the sidearm of the vacuum flask to a vacuum line using thick rubber tubing (see Figure 1). Place a round piece of filter paper in the funnel and dampen it so that it sticks to the plate of the funnel and covers the holes. Turn on the vacuum line, and water and air will be drawn through the filter paper into the flask. At this point, the procedure closely follows the gravity filtration procedure you followed when using filter paper in a conical funnel.

Figure 1. Büchner funnel atop vacuum flask. Be sure to clamp!

Filter the suspension by pouring the supernatant liquid down a glass rod held vertically with the lower end resting against the center of the filter paper. Pouring in this way helps to prevent splashing. However, be careful not to punch a hole in the filter paper with the stirring rod. Be patient while filtering. Do not pour too much liquid into the filter paper. The liquid and the precipitate should always be lower than 1/2 of the way to the top rim of the funnel. Keep the precipitate in the beaker as long as possible, and then pour the suspension down the stirring rod and into the filter. To remove impurities, be sure to wash the precipitate several times with solvent.

PRELAB: PREPARATORY STOICHIOMETRY

Your laboratory procedure will be different from your classmates', depending on your laboratory desk number. Print out and complete the prelab assignment (associated with your laboratory desk number) before coming to the laboratory for this experiment. For instance, if your laboratory desk is 20, then you will print out pages 1-3, 6 & 7 (prelab for desk 20), and 10-14. If desk 13, then you will print out pages 1-3, 4 & 5 (prelab for desk 13), and 10-14.

Prelab for desk numbers 1, 4, 7, 10, 13, 16, 19, and 22

Student Name: ________________________________ *Lab Room No.* _____ *Desk No.* _____

Lab Day __________________________ *Lab Start Time* ________ *Date* __________________

Answer the following questions. Show all work and calculations to support your answers. Include the correct units and the proper number of significant figures on numerical answers. Circle your final answers.

1. For the metal salt indicated in the procedure, you should use $FeCl_2{\cdot}4H_2O$. Calculate the molar mass of $FeCl_2{\cdot}4H_2O$.

2. Your oxalate synthesis product will be $FeC_2O_4{\cdot}2H_2O$. Calculate the molar mass of $FeC_2O_4{\cdot}2H_2O$.

3. Write a balanced molecular equation that shows your metal salt, $FeCl_2{\cdot}4H_2O$, reacting with oxalic acid ($H_2C_2O_4$) to form $FeC_2O_4{\cdot}2H_2O$. What will be the other product(s) of the reaction?

4. Pyrolysis of $FeC_2O_4{\cdot}2H_2O$ will follow one of the following molecular equations. Balance each molecular equation.

A. $FeC_2O_4{\cdot}2H_2O_{(s)} + O_{2(g)} \rightarrow FeO_{(s)} + H_2O_{(g)} + CO_{2(g)}$

B. $FeC_2O_4{\cdot}2H_2O_{(s)} + O_{2(g)} \rightarrow Fe_2O_{3(s)} + H_2O_{(g)} + CO_{2(g)}$

C. $FeC_2O_4{\cdot}2H_2O_{(s)} \rightarrow FeCO_{3(s)} + H_2O_{(g)} + CO_{(g)}$

5. For each of the possible reactions, calculate the theoretical yield (in grams) of the solid product, assuming that you use 1.0 g $FeC_2O_4 \cdot 2H_2O$ and that oxygen is the excess reactant. You will need to write in coefficients from question 4 to balance the molecular equations.

   A. $FeC_2O_4 \cdot 2H_2O_{(s)} + O_{2(g)} \rightarrow FeO_{(s)} + H_2O_{(g)} + CO_{2(g)}$

   B. $FeC_2O_4 \cdot 2H_2O_{(s)} + O_{2(g)} \rightarrow Fe_2O_{3(s)} + H_2O_{(g)} + CO_{2(g)}$

   C. $FeC_2O_4 \cdot 2H_2O_{(s)} \rightarrow FeCO_{3(s)} + H_2O_{(g)} + CO_{(g)}$

Prelab for desk numbers 2, 5, 8, 11, 14, 17, 20, and 23

Student Name: ________________________________ *Lab Room No,* _____ *Desk No.* _____

Lab Day __________________________ *Lab Start Time* ________ *Date* __________________

Answer the following questions. Show all work and calculations to support your answers. Include the correct units and the proper number of significant figures on numerical answers. Circle your final answers.

1. For the metal salt indicated in the procedure, you should use $Ni(NO_3)_2{\cdot}6H_2O$. Calculate the molar mass of $Ni(NO_3)_2{\cdot}6H_2O$.

2. Your oxalate synthesis product will be $NiC_2O_4{\cdot}2H_2O$. Calculate the molar mass of $NiC_2O_4{\cdot}2H_2O$.

3. Write a balanced molecular equation that shows your metal salt, $Ni(NO_3)_2{\cdot}6H_2O$, reacting with oxalic acid ($H_2C_2O_4$) to form $NiC_2O_4{\cdot}2H_2O$. What will be the other product(s) of the reaction?

4. Pyrolysis of $NiC_2O_4{\cdot}2H_2O$ will follow one of the following molecular equations. Balance each molecular equation.

   A. $NiC_2O_4{\cdot}2H_2O_{(s)} + O_{2(g)} \rightarrow NiO_{(s)} + H_2O_{(g)} + CO_{2(g)}$

   B. $NiC_2O_4{\cdot}2H_2O_{(s)} + O_{2(g)} \rightarrow Ni_2O_{3(s)} + H_2O_{(g)} + CO_{2(g)}$

   C. $NiC_2O_4{\cdot}2H_2O_{(s)} \rightarrow NiCO_{3(s)} + H_2O_{(g)} + CO_{(g)}$

5. For each of the possible reactions, calculate the theoretical yield (in grams) of the solid product, assuming that you use 1.0 g $NiC_2O_4 \cdot 2H_2O$ and that oxygen is the excess reactant. You will need to write in coefficients from question 4 to balance the molecular equations.

   A. $NiC_2O_4 \cdot 2H_2O_{(s)} + O_{2(g)} \rightarrow NiO_{(s)} + H_2O_{(g)} + CO_{2(g)}$

   B. $NiC_2O_4 \cdot 2H_2O_{(s)} + O_{2(g)} \rightarrow Ni_2O_{3(s)} + H_2O_{(g)} + CO_{2(g)}$

   C. $NiC_2O_4 \cdot 2H_2O_{(s)} \rightarrow NiCO_{3(s)} + H_2O_{(g)} + CO_{(g)}$

Prelab for desk numbers 3, 6, 9, 12, 15, 18, 21, and 24

Student Name: ________________________________ *Lab Room No.* _____ *Desk No.* _____

Lab Day __________________________ *Lab Start Time* ________ *Date* _________________

Answer the following questions. Show all work and calculations to support your answers. Include the correct units and the proper number of significant figures on numerical answers. Circle your final answers.

1. For the metal salt indicated in the procedure, you should use $MnSO_4{\cdot}H_2O$. Calculate the molar mass of $MnSO_4{\cdot}H_2O$.

2. Your oxalate synthesis product will be $MnC_2O_4{\cdot}3H_2O$. Calculate the molar mass of $MnC_2O_4{\cdot}3H_2O$

3. Write a balanced molecular equation that shows your metal salt, $MnSO_4{\cdot}H_2O$, reacting with oxalic acid ($H_2C_2O_4$) and water to form $MnC_2O_4{\cdot}3H_2O$. What will be the other product(s) of the reaction?

4. Pyrolysis of $MnC_2O_4{\bullet}3H_2O$ will follow one of the following molecular equations. Balance each molecular equation.

A. $MnC_2O_4{\cdot}3H_2O_{(s)} + O_{2(g)} \rightarrow MnO_{(s)} + H_2O_{(g)} + CO_{2(g)}$

B. $MnC_2O_4{\cdot}3H_2O_{(s)} + O_{2(g)} \rightarrow Mn_3O_{4(s)} + H_2O_{(g)} + CO_{2(g)}$

C. $MnC_2O_4{\cdot}3H_2O_{(s)} \rightarrow MnCO_{3(s)} + H_2O_{(g)} + CO_{(g)}$

5. For each of the possible reactions, calculate the theoretical yield (in grams) of the solid product, assuming that you use 1.0 g $MnC_2O_4{\bullet}3H_2O$ and that oxygen is the excess reactant. You will need to write in coefficients from question 4 to balance the molecular equations.

A. $MnC_2O_4{\cdot}3H_2O_{(s)} + O_{2(g)} \rightarrow MnO_{(s)} + H_2O_{(g)} + CO_{2(g)}$

B. $MnC_2O_4{\cdot}3H_2O_{(s)} + O_{2(g)} \rightarrow Mn_3O_{4(s)} + H_2O_{(g)} + CO_{2(g)}$

C. $MnC_2O_4{\cdot}3H_2O_{(s)} \rightarrow MnCO_{3(s)} + H_2O_{(g)} + CO_{(g)}$

PROCEDURE, Part I: SYNTHESIS OF A METAL OXALATE

Have your Prelab approved by your teaching assistant before starting the procedure. Do not work in pairs. All students are to work individually. You will by synthesizing a metal oxalate from a metal salt and oxalic acid via the reaction outlined in Question #3 of your prelab.

1. Weigh between 3.9 g and 4.1 g of your hydrated metal salt (as determined by your desk number: see prelab) into a small beaker using the method of *weighing by difference* (see below). Record the metal salt's exact mass in Table 1. Add 20 mL distilled water and stir to dissolve.
   - To minimize errors inherent in the balance, always use the same balance when weighing.
   - *Weighing by difference:* On an electronic balance, obtain the mass of the clean, dry, empty beaker (to three places after the decimal!). Take this beaker to the triple beam balance and add 3.9 to 4.1 g of metal salt. Return to the same electronic balance and obtain the mass of the beaker plus chemical (to three places after the decimal!). Record this mass and subtract to obtain the mass of metal salt.

   Mass of Beaker + Metal Salt = ________________
   Mass of Empty Beaker = ________________
   Mass of Hydrated Metal Salt = ________________

2. Use a graduated cylinder to precisely measure 50.0 mL of oxalic acid solution. Place the oxalic acid solution in a separate beaker. The solution contains 4.0 g oxalic acid ($H_2C_2O_4$) per 50.0 mL of solution.

3. Once the metal salt is completely dissolved, slowly pour the metal salt solution into the oxalic acid solution. Stir to mix and allow the solution to sit for 10 minutes while a precipitate forms. Try not to remove any precipitate with the stir bar.

4. Prepare a Büchner funnel (as explained in the Scientific Background) while the precipitate is forming. Before placing the filter paper in the Büchner funnel, be sure to weigh and record the mass of your filter paper in Table 1.

5. After the waiting period, begin suction through the filter paper in the funnel and slowly pour your solution onto the center of the filter paper. It is important to try and keep the majority of the precipitate in the center of the paper and not sticking to the funnel walls. The precipitate should be retained on the filter paper as the aqueous solution is drawn into the flask below.

6. Once the entire solution has been emptied into the funnel, add 20 mL of distilled water to the beaker, swirl the beaker to suspend any remaining precipitate, and pour this also into the funnel. You must transfer all of the precipitate from the beaker. Do this as often as needed to get all the precipitate into the funnel.

7. The precipitate will still be wet with water, so you will need to wash the precipitate with acetone. You can do this while the vacuum is still running.

8. Obtain a red-capped acetone wash bottle from under the hood. Gently squirt 5–10 mL of acetone over the precipitate in the funnel. Use the pressure from the wash bottle to focus the acetone stream on the sides of the funnel thus forcing more of the solid back towards the center of the paper. DO NOT add the acetone too quickly or you risk displacing the filter paper and losing sample. Using a glass rod, carefully stir the solid so that all of it is in contact with the acetone. Make sure any solid sticking to the stir rod ends up back on the filter paper (use the acetone wash bottle). During this time, any water on the precipitate particles will dissolve in the acetone and facilitate the drying of your sample.
   - **Acetone is flammable! Keep this liquid away from open flames.**

9. Repeat step 8 with two additional washes of 5–10 mL of acetone.

10. Keep the vacuum on to draw air through the filter paper. The acetone will evaporate from the solid. Keep the vacuum on until the solid is dry. You can push around the precipitate with a glass rod to test for dryness. Once the solid is completely dry, turn off the vacuum.
    - **Be patient!** The drier your solid product, the more accurate your results.
    - This solid product must be **completely dry** before beginning Part II of the procedure.

11. Obtain a plastic weigh dish, find its mass on an electronic balance and record its mass in Table 1. Transfer the dry solid to the weigh dish and determine the mass of the dish plus contents. It may be easier to transfer the entire filter paper and just subtract out the paper's weight. Subtract to obtain the *actual yield* of hydrated metal oxalate precipitate in Table 1.

mass of weigh dish + contents	=	________________
mass of empty weigh dish	=	________________
mass of filter paper	=	________________
actual yield of hyd. metal oxalate ppt.	=	________________
color of hydrated metal oxalate	=	________________

12. Using the data collected (e.g. masses of hydrated metal salt and oxalic acid reacted and actual yield of hydrated metal oxalate precipitate) and the balanced reaction for the formation of the metal oxalate product (see prelab question #3), determine a) the identity of the limiting reactant, b) theoretical yield of the hydrated metal oxalate, and c) the percent yield of the hydrated metal oxalate product. Complete Table 1.

    Calculations: Show your calculations below. Watch your units and report all answers with the correct number of significant figures.

What is pyrolysis?

Pyrolysis is the process of causing a chemical reaction to occur by heating a substance. If there is air present, then pyrolysis may also involve reactions with O_2 or N_2.

PROCEDURE, Part II: PYROLYSIS OF A METAL OXALATE

You will be pyrolyzing the metal oxalate you synthesized in Part I. You will use stoichiometric considerations to determine which pyrolysis reaction (of those given in prelab Question #4) your metal oxalate undergoes.

1. Using the process of *weighing by difference* (see Procedure, Part 1 and Step 1), precisely measure between 0.990 g and 1.010 g of your dry, hydrated metal oxalate into an **aluminum** dish. Record precise masses in Table 2.
   - To enhance heat transfer, spread out the solid over the bottom of the dish.
   - It is important to use between 0.990 g and 1.010 g because your prelab calculations were based upon pyrolysis of 1.0 g of hydrated metal oxalate.
   - Pouring of chemicals should be done at the laboratory bench or on the triple beam balance and not on the electronic balance or in the balance room.
   - Masses obtained using the electronic balance should be given with three decimal places.

   mass of alum. dish + plus hyd. metal oxalate = ________________
   mass of empty aluminum dish = ________________
   mass of hyd. metal oxalate before pyrolysis = ________________

2. Turn the dial on a hot plate to one-half of the maximum setting. Place the aluminum dish on the hot plate. The hydrated metal oxalate will undergo dehydration (loss of water) followed by a pyrolysis reaction once it reaches a certain temperature.

3. Wait until the metal oxalate has been heated at least 10 minutes. Remove the aluminum dish from the hot plate using a pair of metal tongs and place it on the laboratory bench to cool to room temperature. The aluminum dish should cool quickly but the precipitate may take longer.
   - **Caution: the aluminum dish will be VERY hot……….make sure to use the tongs!!**
   - **Be patient!** The cooler the precipitate, the better your results.
   - Never weigh hot objects. Hot objects weigh less than cool objects due to convection currents.

4. Weigh the cooled aluminum dish plus contents and record its mass.

   After first heating: mass of aluminum dish plus contents = _______________

5. Place the aluminum dish back on the hot plate, turn the heat to ~ 400°C, and heat for another 10 minutes, cool to room temperature, and reweigh. If the mass of the dish plus contents has not changed by more than 10 mg (0.010 g), then record the final mass of the dish and product in Table 2. If the mass of the dish plus contents has changed by more than 10 mg (decreased

or increased) then increase the heat (go to a higher temperature, perhaps 450˚C) and heat again for another 10 minutes, cool to room temperature, and reweigh. Repeat this process until the mass remains constant.

- This process of heating, cooling, and reweighing the substance until constant mass is achieved is referred to as *"bringing to constant mass"*.
- Your pyrolysis product should be homogeneous in terms of color. If your pyrolysis product has chunks of yellow intermixed with chunks of black, then pyrolysis is not complete.
- If you are working with the nickel oxalate, do NOT stop the pyrolysis at the intermediate yellow colored substance but rather continue on until you obtain a black substance.

After second heating: mass of aluminum dish plus contents = _______________
After third heating: mass of aluminum dish plus contents = _______________
After fourth heating: mass of aluminum dish plus contents = _______________

6. Complete Table 2. The pyrolysis reaction occurs with near 100% yield, so the actual yield of the pyrolysis reaction should be very close to the theoretical yield you calculated for the correct reaction equation (as calculated in Question #5 of the prelab).

After pyrolysis: final mass of dish + plus pyrolysis product = ________________
mass of empty aluminum dish = ________________
actual mass of pyrolysis product = ________________

color of pyrolysis product = ________________

7. Double check all of your calculations and significant figures. Include units on all of your numbers. Have your teaching assistant verify, by signing, your completion of this experiment. Rip off and hand in the data report sheet (last page) for grading. Dispose of all chemical wastes as indicated by your teaching assistant.

Which of three possible pyrolysis reactions (reaction A, B, or C) actually occurred? _______
Chemical formula of the pyrolysis product from reaction A, B, or C? __________________

First Term General Chemistry
Experiment: Oxalate Stoichiometry

Student Name: ________________________________ *Lab Room No.* _____ *Desk No.* _____
Lab Day ___________________________ *Lab Start Time* ________ *Date* _________________

DATA REPORT SHEET

Table 1. Synthesis of Metal Oxalate

chemical formula of your hydrated metal salt:	
mass of hydrated metal salt used (g)	
volume of oxalic acid solution used (mL)	
mass of oxalic acid (g)	
mass of weigh dish plus contents (g)	
mass of empty weigh dish (g)	
mass of filter paper (g)	
actual yield of hydrated metal oxalate precipitate (g) and its color: ____________________________	
limiting reactant (either hydrated metal salt or oxalic acid)?	
theoretical yield of hydrated metal oxalate product (g)	
percent yield of hydrated metal oxalate product	

Table 2. Pyrolysis of Metal Oxalate

Before pyrolysis: mass of dish plus hydrated metal oxalate (g)	
mass of empty aluminum dish (g)	
Before pyrolysis: mass of hydrated metal oxalate (g)	
After pyrolysis: final mass of dish plus pyrolysis product (g)	
After pyrolysis: actual mass of pyrolysis product (g) and its color: ____________________________	
From prelab (quest. #5): theoretical mass of product in rxn. A (g)	
From prelab (quest. #5): theoretical mass of product in rxn. B (g)	
From prelab (quest. #5): theoretical mass of product in rxn. C (g)	
Using the above information, decide which of the three possible pyrolysis reactions actually occurred, rxn A, B or C?	
Chemical formula of the metal containing pyrolysis product from rxn. A, B, or C?	

Teaching Assistant Use Only **Experiment Grade** _________%

Did the student have the pre-lab exercises completed before lab? Yes ____ No____
The teaching assistant will sign below once satisfied that the student has performed the entire procedure. The report will not be accepted or graded unless signed.

Teaching Assistant's Approval: ____________________________________
Signature

Oxalate Stoichiometry
Instructor Notes

Safety Precautions

- Students should be required to wear chemical splash goggles, gloves, and apron at all times.
- Caution students not to touch hot objects. Immediately place any burns under cool running water and inform the teaching assistant.
- Have students perform the steps involving heating and pyrolysis of the solid oxalate well inside the student hood.
- Most metal salts are toxic if ingested. Some, like $Ni(NO_3)_2{\bullet}6H_2O$, are carcinogens. Students should be required to wear gloves!
- Students should wash hands with soap and water after completing the experiment and before leaving the laboratory room.

Chemicals Needed

- 0.888 M oxalic acid, $H_2C_2O_4$ prepared from oxalic acid dihydrate, $H_2C_2O_4{\bullet}2H_2O$ (CAS registry number 6153-56-6)
- iron(II) chloride tetrahydrate, $FeCl_2{\bullet}4H_2O$ (CAS registry number 13478-10-9)
- nickel(II) nitrate hexahydrate, $Ni(NO_3)_2{\bullet}6H_2O$ (CAS registry number 13478-00-7)
- manganese(II) sulfate monohydrate, $MnSO_4{\bullet}H_2O$ (CAS registry number 10034-96-5)
- acetone, C_3H_6O (CAS registry number 665-52-4)

Chemical Preparation

- To prepare the oxalic acid solution, dissolve 112 g of oxalic acid dihydrate in one liter of distilled water.
- Put the acetone out in red labeled/capped safety-vented low-density polyethylene wash bottles.

Equipment Needed

- aluminum weigh dishes (1 per student) for heating of the oxalate
- plastic weigh dishes
- stir bars and stirring hot plates with ceramic tops that heat to 500˚C ─ we used Fisher Scientific Isotemp stirring hot plates (Cat. No. 11-100-16SH) with a temperature readout to monitor pyrolysis temperature
- Büchner funnels, with associated heavy walled hoses and filter paper
- metal tongs, tweezers, or forceps for removal of the hot aluminum weigh dish

Other Notes

- During pyrolysis, the solid form is maintained and splattering does not occur.
- To enhance heat transfer during pyrolysis, remind the students to spread out the solid over the bottom of the aluminum dish.
- For more advanced students, this experiment can be altered to require students to identify, for themselves, chemical formulas for 1) the hydrated metal oxalate and 2) its pyrolysis product using experimental data obtained from TGA, XPS, and

XRD analyses. Students would then discover the mixed nature of the manganese oxalate's pyrolysis product.

Color Changes Upon Pyrolysis

- iron(II) oxalate dihydrate: yellow to red-brown or rust colored Fe_2O_3 product
- nickel(II) oxalate dihydrate: bluish-green to black NiO product (Caution students NOT to stop at the intermediate yellow colored substance!)
- manganese(II) oxalate trihydrate: light pink to brown-black Mn_3O_4 product (actually a mixture of Mn_3O_4, major product, and Mn_2O_3, minor product, but both products give ~ same mass upon pyrolysis of 1.0 g of oxalate and both have similar colors.)

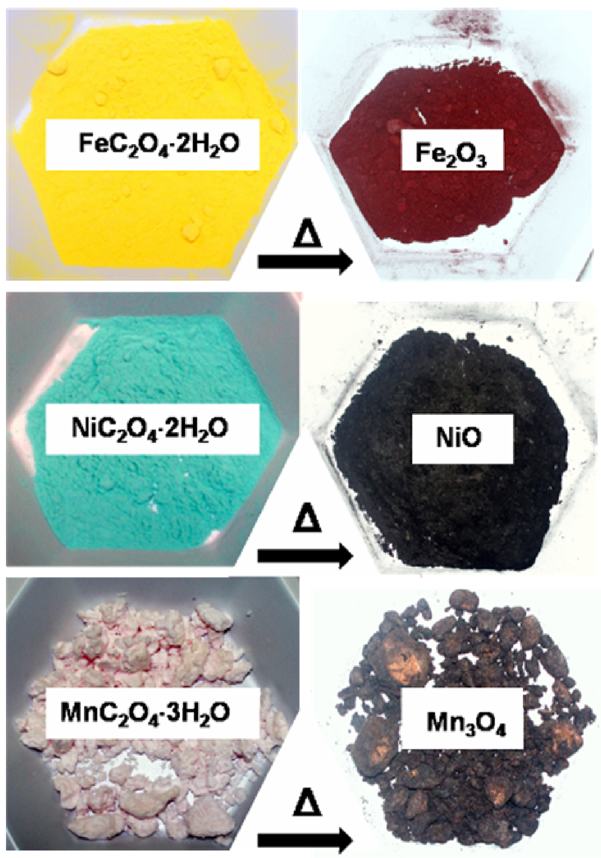

Figure 1: Colors of the Hydrated Metal Oxalates and their Pyrolysis Products

Answer Keys to Prelaboratory Questions

Prelab for desk numbers 1, 4, 7, 10, 13, 16, 19, and 22

1. For the metal salt indicated in the procedure, you should use $FeCl_2 \cdot 4H_2O$. Calculate the molar mass of $FeCl_2 \cdot 4H_2O$. **Answer: Molar Mass = 198.81 g mol^{-1}**

2. Your oxalate synthesis product will be $FeC_2O_4 \cdot 2H_2O$. Calculate the molar mass of $FeC_2O_4 \cdot 2H_2O$. **Answer: Molar Mass = 179.90 g mol^{-1}**

3. Write a balanced molecular equation that shows your metal salt, $FeCl_2 \cdot 4H_2O$, reacting with oxalic acid ($H_2C_2O_4$) to form $FeC_2O_4 \cdot 2H_2O$. What will be the other product(s) of the reaction?
Answer: $FeCl_2 \bullet 4H_2O(aq) + H_2C_2O_4(aq) \rightarrow FeC_2O_4 \bullet 2H_2O(s) + 2H_2O(l) + 2\ HCl(aq)$

4. Pyrolysis of $FeC_2O_4 \cdot 2H_2O$ will follow one of the following molecular equations. Balance each molecular equation.
Answer:
A. **$2\ FeC_2O_4 \cdot 2H_2O_{(s)} + O_{2(g)} \rightarrow 2\ FeO_{(s)} + 4\ H_2O_{(g)} + 4\ CO_{2(g)}$**
B. **$4\ FeC_2O_4 \cdot 2H_2O_{(s)} + 3\ O_{2(g)} \rightarrow 2\ Fe_2O_{3(s)} + 8\ H_2O_{(g)} + 8\ CO_{2(g)}$**
C. **$FeC_2O_4 \cdot 2H_2O_{(s)} \rightarrow FeCO_{3(s)} + 2\ H_2O_{(g)} + CO_{(g)}$**

5. For each of the possible reactions, calculate the theoretical yield (in grams) of the solid product, assuming that you use 1.0 g $FeC_2O_4 \cdot 2H_2O$ and that oxygen is the excess reactant. You will need to write in coefficients from question 4 to balance the molecular equations.
Answer:
A. $2\ FeC_2O_4 \cdot 2H_2O_{(s)} + O_{2(g)} \rightarrow 2\ FeO_{(s)} + 4\ H_2O_{(g)} + 4\ CO_{2(g)}$

$$1.0\text{ g} \times \frac{1\text{ mol } FeC_2O_4 \cdot 2H_2O}{179.90\text{ g}} \times \frac{2\text{ mol FeO}}{2\text{ mol } FeC_2O_4 \cdot 2H_2O} \times \frac{71.85\text{ g FeO}}{1\text{ mol FeO}} = \mathbf{0.40\ g\ FeO\ (2\ sf)}$$

B. $4\ FeC_2O_4 \cdot 2H_2O_{(s)} + 3\ O_{2(g)} \rightarrow 2\ Fe_2O_{3(s)} + 8\ H_2O_{(g)} + 8\ CO_{2(g)}$

$$1.0\text{ g} \times \frac{1\text{ mol } FeC_2O_4 \cdot 2H_2O}{179.90\text{ g}} \times \frac{2\text{ mol } Fe_2O_3}{4\text{ mol } FeC_2O_4 \cdot 2H_2O} \times \frac{159.69\text{ g } Fe_2O_3}{1\text{ mol } Fe_2O_3} = \mathbf{0.44\ g\ Fe_2O_3\ (2\ sf)}$$

C. $FeC_2O_4 \cdot 2H_2O_{(s)} \rightarrow FeCO_{3(s)} + 2\ H_2O_{(g)} + CO_{(g)}$

$$1.0\text{ g} \times \frac{1\text{ mol } FeC_2O_4 \cdot 2H_2O}{179.90\text{ g}} \times \frac{1\text{ mol } FeCO_3}{1\text{ mol } FeC_2O_4 \cdot 2H_2O} \times \frac{115.86\text{ g } FeCO_3}{1\text{ mol } FeCO_3} = \mathbf{0.64\ g\ FeCO_3\ (2\ sf)}$$

Answer Keys to Prelaboratory Questions

Prelab for desk numbers 2, 5, 8, 11, 14, 17, 20, and 23

1. For the metal salt indicated in the procedure, you should use $Ni(NO_3)_2{\cdot}6H_2O$. Calculate the molar mass of $Ni(NO_3)_2{\cdot}6H_2O$. **Answer: Molar Mass = 290.80 g mol^{-1}**

2. Your oxalate synthesis product will be $NiC_2O_4{\cdot}2H_2O$. Calculate the molar mass of $NiC_2O_4{\cdot}2H_2O$. **Answer: Molar Mass = 182.76 g mol^{-1}**

3. Write a balanced molecular equation that shows your metal salt, $Ni(NO_3)_2{\cdot}6H_2O$, reacting with oxalic acid ($H_2C_2O_4$) to form $NiC_2O_4{\cdot}2H_2O$. What will be the other product(s) of the reaction?
Answer: $Ni(NO_3)_2{\bullet}6H_2O(aq) + H_2C_2O_4(aq) \rightarrow NiC_2O_4{\bullet}2H_2O(s) + 2HNO_3(aq) + 4H_2O(l)$

4. Pyrolysis of $NiC_2O_4{\cdot}2H_2O$ will follow one of the following molecular equations. Balance each molecular equation.
Answer:
A. **$2\ NiC_2O_4{\cdot}2H_2O_{(s)} + O_{2(g)} \rightarrow 2\ NiO_{(s)} + 4\ H_2O_{(g)} + 4\ CO_{2(g)}$**
B. **$4\ NiC_2O_4{\cdot}2H_2O_{(s)} + 3\ O_{2(g)} \rightarrow 2\ Ni_2O_{3(s)} + 8\ H_2O_{(g)} + 8\ CO_{2(g)}$**
C. **$NiC_2O_4{\cdot}2H_2O_{(s)} \rightarrow NiCO_{3(s)} + 2\ H_2O_{(g)} + CO_{(g)}$**

5. For each of the possible reactions, calculate the theoretical yield (in grams) of the solid product, assuming that you use 1.0 g $NiC_2O_4{\cdot}2H_2O$ and that oxygen is the excess reactant. You will need to write in coefficients from question 4 to balance the molecular equations.
Answer:
A. $2\ NiC_2O_4{\cdot}2H_2O_{(s)} + O_{2(g)} \rightarrow 2\ NiO_{(s)} + 4\ H_2O_{(g)} + 4\ CO_{2(g)}$

$$1.0\ \text{g} \times \frac{1\ \text{mol}\ NiC_2O_4{\cdot}2H_2O}{182.76\ \text{g}} \times \frac{2\ \text{mol NiO}}{2\ \text{mol}\ NiC_2O_4{\cdot}2H_2O} \times \frac{74.71\ \text{g NiO}}{1\ \text{mol NiO}} = \mathbf{0.41\ g\ NiO\ (2\ sf)}$$

B. $4\ NiC_2O_4{\cdot}2H_2O_{(s)} + 3\ O_{2(g)} \rightarrow 2\ Ni_2O_{3(s)} + 8\ H_2O_{(g)} + 8\ CO_{2(g)}$

$$1.0\ \text{g} \times \frac{1\ \text{mol}\ NiC_2O_4{\cdot}2H_2O}{182.76\ \text{g}} \times \frac{2\ \text{mol}\ Ni_2O_3}{4\ \text{mol}\ NiC_2O_4{\cdot}2H_2O} \times \frac{165.38\ \text{g}\ Ni_2O_3}{1\ \text{mol}\ Ni_2O_3} = \mathbf{0.45\ g\ Ni_2O_3\ (2\ sf)}$$

C. $NiC_2O_4{\cdot}2H_2O_{(s)} \rightarrow NiCO_{3(s)} + 2\ H_2O_{(g)} + CO_{(g)}$

$$1.0\ \text{g} \times \frac{1\ \text{mol}\ NiC_2O_4{\cdot}2H_2O}{182.76\ \text{g}} \times \frac{1\ \text{mol}\ NiCO_3}{1\ \text{mol}\ NiC_2O_4{\cdot}2H_2O} \times \frac{118.72\ \text{g}\ NiCO_3}{1\ \text{mol}\ NiCO_3} = \mathbf{0.65\ g\ NiCO_3\ (2\ sf)}$$

Answer Keys to Prelaboratory Questions

Prelab for desk numbers 3, 6, 9, 12, 15, 18, 21, and 24

1. For the metal salt indicated in the procedure, you should use $MnSO_4 \cdot H_2O$. Calculate the molar mass of $MnSO_4 \cdot H_2O$. **Answer: Molar Mass = 169.02 g mol^{-1}**

2. Your oxalate synthesis product will be $MnC_2O_4 \cdot 3H_2O$. Calculate the molar mass of $MnC_2O_4 \cdot 3H_2O$ **Answer: Molar Mass = 197.01 g mol^{-1}**

3. Write a balanced molecular equation that shows your metal salt, $MnSO_4 \cdot H_2O$, reacting with oxalic acid ($H_2C_2O_4$) and water to form $MnC_2O_4 \cdot 3H_2O$. What will be the other product(s) of the reaction?
Answer: $MnSO_4 \bullet H_2O(aq) + H_2C_2O_4(aq) + 2\ H_2O(l) \rightarrow MnC_2O_4 \bullet 3H_2O(s) + H_2SO_4(aq)$

4. Pyrolysis of $MnC_2O_4 \bullet 3H_2O$ will follow one of the following molecular equations. Balance each molecular equation.
Answer:
A. $2\ MnC_2O_4 \cdot 3H_2O_{(s)} + O_{2(g)} \rightarrow 2\ MnO_{(s)} + 6\ H_2O_{(g)} + 4\ CO_{2(g)}$
B. $3\ MnC_2O_4 \cdot 3H_2O_{(s)} + 2\ O_{2(g)} \rightarrow Mn_3O_{4(s)} + 9\ H_2O_{(g)} + 6\ CO_{2(g)}$
C. $MnC_2O_4 \cdot 3H_2O_{(s)} \rightarrow MnCO_{3(s)} + 3\ H_2O_{(g)} + CO_{(g)}$

5. For each of the possible reactions, calculate the theoretical yield (in grams) of the solid product, assuming that you use 1.0 g $MnC_2O_4 \bullet 3H_2O$ and that oxygen is the excess reactant. You will need to write in coefficients from question 4 to balance the molecular equations.
Answer:
A. $2\ MnC_2O_4 \cdot 3H_2O_{(s)} + O_{2(g)} \rightarrow 2\ MnO_{(s)} + 6\ H_2O_{(g)} + 4\ CO_{2(g)}$

$$1.0\text{ g} \times \frac{1\text{ mol } MnC_2O_4 \cdot 3H_2O}{197.01\text{ g}} \times \frac{2\text{ mol } MnO}{2\text{ mol } MnC_2O_4 \cdot 3H_2O} \times \frac{70.94\text{ g } MnO}{1\text{ mol } MnO} = \mathbf{0.36\ g\ MnO\ (2\ sf)}$$

B. $3\ MnC_2O_4 \cdot 3H_2O_{(s)} + 2\ O_{2(g)} \rightarrow Mn_3O_{4(s)} + 9\ H_2O_{(g)} + 6\ CO_{2(g)}$

$$1.0\text{ g} \times \frac{1\text{ mol } MnC_2O_4 \cdot 3H_2O}{197.01\text{ g}} \times \frac{1\text{ mol } Mn_3O_4}{3\text{ mol } MnC_2O_4 \cdot 3H_2O} \times \frac{228.81\text{ g } Mn_3O_4}{1\text{ mol } Mn_3O_4} = \mathbf{0.39\ g\ Mn_3O_4\ (2\ sf)}$$

C. $MnC_2O_4 \cdot 3H_2O_{(s)} \rightarrow MnCO_{3(s)} + 3\ H_2O_{(g)} + CO_{(g)}$

$$1.0\text{ g} \times \frac{1\text{ mol } MnC_2O_4 \cdot 3H_2O}{197.01\text{ g}} \times \frac{1\text{ mol } MnCO_3}{1\text{ mol } MnC_2O_4 \cdot 3H_2O} \times \frac{114.95\text{ g } MnCO_3}{1\text{ mol } MnCO_3} = \mathbf{0.58\ g\ MnCO_3\ (2\ sf)}$$

Oxalate Stoichiometry

Grading Key

<u>Experiment: Oxalate Stoichiometry</u>

If no teaching assistant signature on last page, do not grade but bring to attention of lab room staff who will then bring it to the attention of the instructor.

Assign grades based on the following:

1.	Did student complete the entire experiment?	YES = 60%	NO = 0%
2.	Did student have the prelab questions completed prior to coming to the laboratory?	YES = 0%	NO = −20%
3.	Did student use the correct number of significant figures? (do not compound -10 for one or all sig fig errors)	YES = 0%	NO = −10%
4.	Did student have a subtraction error in calculating actual mass of pyrolysis product (see Table 2 5th entry)?	YES = −10%	NO = 0%
5.	Did student work in a group? (they were to work individually on this experiment)	YES = −10%	NO = 0%

6. Did student have the correct answers to six of the questions in Tables 1 and 2 of the Data Report Sheet as specified in the following table?

Metal Salt Starting Material	$FeCl_2 \cdot 4\ H_2O$	$Ni(NO_3)_2 \cdot 6\ H_2O$	$MnSO_4 \cdot H_2O$
Table 1 (9th entry): Limiting reactant? (either metal salt or oxalic acid)	metal salt (+5 pts)	metal salt (+5 pts)	metal salt (+5 pts)
Table 1 (10th entry): Theoretical yield of metal oxalate product. You will need to calculate and compare based upon mass of metal oxalate salt used (2nd entry in Table 1). Answer should have 4 sf.	mass metal oxalate used (2nd entry) × 0.9048 = (+5 pts; 4 sf)	mass metal oxalate used (2nd entry) × 0.6284 = (+5 pts; 4 sf)	mass metal oxalate used (2nd entry) × 1.165 = (+5 pts; 4 sf)
Table 2 (6th entry): From prelab (quest #5): theoretical mass of product in rxn A (2 sf)	0.40 g (+5 pts; 2 sf)	0.41 g (+5 pts; 2 sf)	0.36 g (+5 pts; 2 sf)
Table 2 (7th entry): From prelab (quest #5): theoretical mass of product in rxn A (2 sf)	0.44 g (+5 pts; 2 sf)	0.45 g (+5 pts; 2 sf)	0.39 g (+5 pts; 2 sf)
Table 2 (8th entry): From prelab (quest #5): theoretical mass of product in rxn A (2 sf)	0.64 g (+5 pts; 2 sf)	0.65 g (+5 pts; 2 sf)	0.58 g (+5 pts; 2 sf)
Table 2 (9th entry): Decide which of the three possible pyrolysis reactions actually occurred, rxn A, B, or C?	B (+15 pts)	A (+15 pts)	B (+15 pts)

It will be easier and less time consuming to grade all report sheets for students who used the iron starting material first, followed by the nickel, and the manganese last. In this way, you begin to memorize the numbers involved and don't have to continually refer back to the answer key. Add up the percentages and assign a grade out of 100 %.

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